

JOY SAHA

Albany, NY-12222, United States

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Education

University At Albany, SUNY

PhD in Electrical and Computer Engineering

Major in Signal Processing and Communications

Advisor: Professor Daphney-Stavroula Zois

Aug 2024 – Present

NY, United States

CGPA: 4.00/4.00

Bangladesh University of Engineering and Technology

Bachelor of Science in Electrical and Electronic Engineering

Major in Communication and Signal Processing

Achieved an average CGPA of 3.75 across the last four terms.

Mar 2018 – May 2023

Dhaka, Bangladesh

CGPA: 3.61/4.00

Research Interests

- Costly Feature Acquisition
- Multi-view Learning
- Statistical Signal Processing
- Sequential Decision Making
- Optimization
- Instant-wise Feature Grouping

Experience

Graduate Teaching Assistant

University at Albany, SUNY

Department of Electrical and Computer Engineering

Aug 2024 – Present

NY, United States

Research Project Assistant

The Research Foundation for SUNY

Department of Electrical and Computer Engineering

Summer 2025, 2026

NY, United States

Part-time Lecturer

Presidency University

Department of Electrical and Electronic Engineering

Jan 2024 – Aug 2024

Dhaka, Bangladesh

Adjunct Lecturer

Bangladesh University of Textiles

Department of Textile Machinery Design and Maintenance

Jan 2024 – July 2024

Dhaka, Bangladesh

Publications

- **Joy Saha**, Daphney-Stavroula Zois. “Adaptive Sequential Feature Grouping and Acquisition for Cost-Efficient Classification.” 34th European Signal Processing Conference (EUSIPCO), 2026. (Accepted)
- Lamia Hossain, Ilma Hossain, SM Taslim Uddin Raju, Md Shahidul Salim, and **Joy Saha**, “A Novel Technique for Classification of Motor Imagery EEG Signal Based on Deep Learning Approaches,” *Intelligent Networks and Systems*, pp. 52–64, Chapman and Hall/CRC, 2025.

Projects

CSI-Based Sign Language Recognition Using CNN-GRU Architecture Enhanced with Attention | [GitHub](#)

- Designed a camera-free WiFi CSI-based sign language recognition system using CNN for spatial feature extraction across subcarriers and antennas.
- Developed a Bi-GRU with attention mechanism for temporal modeling, combined with self-supervised contrastive pretraining and extensive CSI augmentation.
- Achieved up to 99% accuracy, validated through ablation studies and Grad-CAM interpretability analysis using deep sequential models and attention mechanisms.

Neural Networks: Investigating Effects of Layer Depth, Neuron Count, and Activation Functions | [GitHub](#)

- Implemented feedforward neural networks from scratch in Python, systematically analyzing the impact of layer depth and neuron count on classification performance and generalization.

- Investigated the effect of various activation functions on computational efficiency across standard datasets using back-propagation and optimization techniques.
- Built intuition about data transformations through learned weights and activations, evaluating trade-offs between model complexity and performance.

Comparative Analysis of Ada-Hessian and First-Order Optimizers for CSI-Based Sign Language Recognition | [GitHub](#)

- Conducted a comparative study of the Ada-Hessian second-order optimizer against first-order methods including SGD, Adam, AdamW, Adamax, Nadam, and RMSprop using stochastic Hessian estimation.
- Demonstrated faster convergence (≈ 50 vs. $150+$ epochs) and superior validation accuracy of 99.31% on the Lab dataset for WiFi CSI-based sign language recognition.
- Analyzed robustness to learning-rate variations and per-iteration computational cost trade-offs across all optimizers in a deep learning framework.

Social Distance Monitoring in COVID-19 Situation | [GitHub](#)

- Employed machine learning techniques using Matlab, CNN, and YOLOv2 to create a system that detects social distance compliance in images or videos.
- Trained models using annotated datasets to classify individuals and visually indicate their proximity to others with green and red bounding boxes.
- Implemented the system as a DSP lab project using image processing techniques, enabling real-time monitoring and visualization of social distancing.

Bayesian Hypothesis Testing on SeeClickFix Service Requests | [GitHub](#)

- Designed Bayes-optimal binary and 4-hypothesis tests using Bayes criterion and likelihood ratio testing to classify 529 citizen 311 service requests from the Albany County SeeClickFix dataset.
- Applied multi-variate Bernoulli and multinomial likelihood models with Laplace-smoothed maximum-likelihood estimates, evaluating performance metrics (P_F , P_D , P_M , P_e) across a grid of Bayes cost tuples.
- Traced ROC curves achieving AUC up to 0.995 with test errors as low as 1.9%, demonstrating strong classification performance.

IoT Based Environment Monitor System | [GitHub](#)

- Developed an IoT-based environment monitoring system using Arduino, NodeMCU, MQ135, DHT11, SIM800L, and LCD in the Microprocessor and Embedded System sessional course, incorporating GSM technology for seamless connectivity.
- Utilized two sensors to measure temperature, humidity, and air quality, with the collected data uploaded to an online cloud system for real-time display and recording.
- Implemented a warning mechanism where users receive immediate notifications on their phones when the monitored parameters exceed predetermined thresholds.
- Powered the entire system using solar energy with battery backup, ensuring reliable operation and sustainability.

IoT Based Remote Heart Rate Monitoring System | [GitHub](#)

- Developed an IoT-based system using Arduino, ESP8266, and AD8232 for collecting ECG data from patients, utilizing a microcontroller for heart rate calculation and identification of abnormal ECG signals.
- Implemented email notifications and data uploading to Google Drive for further analysis, enabling efficient communication and storage of the collected ECG data.
- Designed the system to integrate IoT technology, microcontroller-based calculations, and data management functionalities for enhanced monitoring and analysis of ECG signals.

Undergraduate Thesis

Implementation-Friendly Convolution Neural Network for Sign Language Recognition Using WiFi CSI Data | [GitHub](#)

- Collaborated with Dr. Hafiz Imtiaz and Dr. Tahsina Farah Sanam to develop an innovative WiFi CSI-based sign language recognition system using Matlab and Python as an undergraduate thesis.
- Employed advanced machine learning algorithms and signal processing techniques to interpret sign language gestures captured through WiFi signals, aiming for accurate recognition.
- Utilized MATLAB for preprocessing the captured CSI data and Python for implementing machine learning models, demonstrating the versatility and effectiveness of the combined approach.
- Achieved comparable accuracy to existing methods while employing a less complex model and simpler preprocessing techniques, indicating the efficiency and effectiveness of the developed system.
- Contributed to the field of sign language recognition by offering a WiFi-based solution with the potential to improve communication and accessibility for individuals using sign language.

Technical Skills

Languages: Python, C, C++, Matlab, Verilog

Simulation Tools: PSpice, TinkerCad, PSAF, PSSE, AutoCad, Proteus

Others: Word, Excel, PowerPoint, Latex, PLC, PCB Design

Relevant Coursework

- | | | |
|----------------------------------|---------------------------------|--|
| • Probabilistic Machine Learning | • Optimization Methods and Non- | • Advanced Linear Algebra |
| • Deep Learning | Linear Programming | • Probability and Random Processes |
| • Algorithms and Data Structures | • Artificial Intelligence I | • Foundations of Statistical Inference |

Standardized Test Scores

GRE General Test: Quant- 165/170, Verbal 148/170, AWA-3.0/6.0

Nov 07,2023

TOEFL iBT Test: Reading- 28/30, Listening- 25/30, Speaking- 22/30, Writing- 27/30

Dec 08,2023

Volunteer Work

Buet Rover Scout

Oct 2018

Volunteer During Covid Pandemic

April 2020 – Aug 2020

Organizer EEE Day

Feb 2023

Awards and Scholarships

CNSE Excellence in Teaching

2025, 2026

- Awarded Best Graduate Teaching Assistant, Department of Electrical and Computer Engineering, UAlbany.

Coursera Financial Aid

2026

- Awarded Financial Aid for Fundamentals of Reinforcement Learning Course.

Dean's List

Feb 2021 - April 2022

- Acknowledged for achieving an average CGPA of 3.75 or above across consecutive terms(3-1, 3-2).

References

Dr. Daphney–Stavroula Zois

Associate Professor, Department of

Electrical & Computer Engineering

University at Albany, SUNY

NY, United States

✉ dzois.albany.edu

☎ On Request

Dr. Hafiz Imtaiz

Professor, Department of

Electrical & Electronic Engineering

Bangladesh University of Engineering and

Technology, Dhaka, Bangladesh

✉ hafizimtiaz@eee.buet.ac.bd

☎ On Request